

Research Article

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Feeling fast? Beliefs and performance among high school sprinters

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Abstract: Mindset can shape sports performance, but these effects can be difficult to detect empirically. We use data from high school sprinters to explore mindset effects on 100 m finishing times and find that headwinds hamper performance more than can be attributed to the physics of wind resistance alone. These (implied) psychological effects of wind on sprint times are stronger for girls than for boys. Having established the presence of mindset-based slack in physical performance, we then test whether sprint times changed in the wake of Matthew Boling's record-breaking sprint in 2019 that, after going viral on social media, potentially boosted self-efficacy among high school sprinters. Using 2018 and 2019 high school track meets in California, we observe notable changes in average sprinter performance for certain types of athletes in specific wind conditions after Boling's race that did not occur in the previous season. These results control for many observable variables, correct for multiple hypothesis testing, and use entropy balancing weights to ensure statistical comparability between the two years. We detect differences in this 'Boling effect' based on the predicted racial composition of teams and the predicted race of athletes, which is relevant given the racial angle of coverage of the record-setting run. We posit vicarious self-efficacy as a plausible explanation for these difference-in-differences patterns. Pronounced heterogeneity in these results demonstrates some of the complexities and nuances of mindset effects even in settings where performance is easily quantified. Subtle contextual and salience cues appear to mediate the impact of vicarious self-efficacy beliefs on performance.

Keywords: performance; self-efficacy; sports psychology; technical efficiency; vicarious effects

1 Introduction

The impact of beliefs on performance and productivity is evident intuitively across many domains and disciplines although empirically isolating and estimating these effects is challenging. In economics, recent work on aspirations and self-efficacy has grappled with how beliefs can alter individual productivity relative to hard external constraints (e.g., Appadurai 2004; Bernard et al. 2014; Dalton, Ghosal, and Mani 2016; Lybbert and Wydick 2018; Wuepper and Lybbert 2017). One common reaction to this research might be called "The Peter Pan Critique": Believing doesn't make it so! While this is certainly true when biophysical constraints bind (e.g., flying in the Peter Pan case), it may be less valid as a concern in contexts where such constraints are partially non-binding. In such cases, beliefs can and often do directly shape effort, productivity and outcomes.

Competitive sports are frequently used as the prototypical case for how mindset can shape performance (e.g., Imperiale-Hagerman 2011; Johnson et al. 2009; Shafizadeh and Gray 2011), but statistically isolating these effects is notoriously difficult. We argue that the 100 meter (100 m) sprint event in high school track and field offers a unique setting for teasing out these psychological effects. A prevailing headwind or tailwind directly changes the wind resistance (i.e., drag) faced by the sprinter, but its effect on finishing times is also a function of indirect psychological effects. For elite world-class sprinters, these indirect mindset effects are largely neutralized by years of intense mental and physical training. Their sprint times in windy conditions therefore reflect predominantly changes in drag with only minimal psychological effects. Elite sprinters therefore provide a useful benchmark for high school sprinters, who—with less experience and training—are subject to both direct physical and indirect psychological wind effects.

We use data from high school track meets in the U.S. to test how wind conditions affect finishing times for boys and girls. We proceed in two empirical parts. First, comparing these effects to elite sprinters, we establish the presence of slack in the performance of high school sprinters: headwinds slow high school sprinters more than wind resistance

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alone affects performance, an effect that is particularly pronounced for girls. This implied evidence of measurable mindset effects opens the door for the second part of our data analysis, which explores the effects of a vicarious self-efficacy boost among these sprinters.

Bandura (1977) first formalized the concept of self-efficacy and triggered a generation of research in psychology on the effects of self-efficacy beliefs on performance. Taken as a whole, this work demonstrates the conditions under which self-efficacy—individuals' perceptions of their ability to achieve a desired outcome—can directly enhance performance. Many of the earliest tests of Bandura (1977) involved physical or athletic tasks. For example, Weinberg et al. (1979) devised a competitive leg-endurance experiment in which they exogenously vary subjects' self-efficacy by leading them to believe they were competing against a varsity track athlete (low self-efficacy treatment) or an individual who recently sustained a knee injury (high self-efficacy treatment) and found that those with experimentally-induced self-efficacy exert greater effort and are more persistent. Bandura (1977) included such vicarious experiences as one of four main sources of self-efficacy and later described "efficacy appraisals [as] partly influenced by vicarious experiences mediated through modeled attainments" (Bandura 1997, p. 86). Feltz et al. (1979) and McAuley (1985) demonstrated how improved self-efficacy based on such modeling can directly enhance sports performance, something effective coaches obviously know and actively leverage.

Instead of experimentally manipulating the self-efficacy of experimental subjects, we use data from high school track meets in California (CA) and leverage an exogenous and vicarious shock to self-efficacy beliefs for our test. Specifically, we consider Matthew Boling's record-setting 9.98 s 100 m sprint on April 27, 2019, video of which went viral among track fans and athletes and was covered by general media outlets as well. His race not only set the record as the all-time fastest "all conditions" 100 m sprint for high school boys, but also made him the youngest sprinter to ever break 10 s.¹ Discussion of this

remarkable performance often emphasized the fact that Boling is white.²

To explore the potential effects of Boling's record-breaking performance on subsequent 100 m times posted by high school sprinters, we use a difference-in-difference (DiD) identification strategy and a statistical approach that incorporates entropy balancing weights (Hainmueller 2012; Hainmueller and Xu 2013) and corrects for multiple hypothesis testing. This empirical approach provides second-best causal inference while allowing for several dimensions of heterogeneity, including gender, wind conditions, and the racial composition of teams, which we predict using a modified version of the Bayesian Improved Firstname Surname Geocoding (BIFSG) algorithm proposed by Voicu (2018). Because we neither manipulate nor measure athlete beliefs, we can only hypothesize the role mindset effects may play as an explanation. Since our first set of results suggest that headwinds introduce psychological effects into sprinter performance, however, this seems plausible.

Our second set of results are consistent with Boling's record triggering complex and distinctly heterogeneous effects on subsequent finishing times. In contrast to a simple vicarious self-efficacy response, we find evidence of a complex Boling effect³ consisting of two opposing impacts on performance: some types of athletes indeed seem to run faster immediately after this exogenous shock, but other types appear to run *slower*. Since slower times are most apparent in tailwind conditions, this might reflect amplified overconfidence due to feeling fast. These opposing effects appear to be moderated importantly by gender and the predicted race of athletes and teams. Overall, the results suggest that in the presence of psychological slack in performance, vicarious self-efficacy beliefs can have measurable impacts on performance, but subtle contextual and salience cues shape these effects.

This paper makes three primary contributions. First, our conceptual approach uses predictions from fluid dynamic models of drag on elite sprinters to benchmark the physical effects of wind on sprint times. This benchmark

1 Dozens of videos of this race circulated on social media, including this one: <https://www.youtube.com/watch?v=ejoAhVDhA5w> (accessed 16 June 2023). The race was run with a brisk tailwind, which is apparent in the linked video when an official holds up a white flag after the conclusion of the race. Because the tailwind was stronger than the allowable 2 m/s, it did not post as an official time but as an "all conditions" time, but this fact did not diminish excitement about and widespread coverage of the feat. Boling subsequently competed at the University of Georgia where he was especially dominant in the 200 m

event in which he was ranked 15th in the world at one point and demonstrated his potential as a world-class sprinter.

2 Boling was the only white sprinter in the final heat when he posted this time. Video footage of the sprint often circulated under the banner "White Lightning," much to the dismay and objections of Boling himself.

3 We use "Boling effect" as a simple shorthand term for the analysis that follows. We use it to refer to the potential impacts that Boling's performance had on subsequent sprint times as mediated by mindset changes. Given the unique confluence of events we study, we do not intend for the term to be generalized beyond the context of this study.

then allows us to identify additional indirect effects of wind that operate through the psychology of amateur sprinters. Second, our modification of the BIFSG algorithm for predicting race of individuals (Voicu 2018) opens new possibilities for empirical analysis, in particular using administrative or online records of elementary or secondary school performance, whether academic or athletic. Third, the results of our analysis indicate a clear role of gender in mindset effects on sprinter performance and suggest that vicarious self-efficacy effects are nuanced and distinctly heterogeneous, but may nonetheless be empirically detectable.

2 Conceptual approach

In this section, we describe two core dimensions of our analytical approach in this paper. In the first, we describe in greater detail the way we parse out the direct effect of wind on sprinter performance via drag and the less direct (but no less real) effect of wind via the psychology of the sprinter. In the second, we discuss potential heterogeneity in vicarious self-efficacy that may shape what we call the “Boling effect”: the impact Boling’s viral and record-breaking 100 m performance had on subsequent high school sprinter times as mediated through self-efficacy beliefs. These conceptual details influence our empirical approach described below.

2.1 Psychology & physics: wind effects on sprinters

Beliefs can only shape performance when there is slack in baseline performance relative to hard biophysical constraints. We establish the presence and magnitude of this slack by leveraging a specific and very salient feature of 100 m sprints: prevailing wind affects performance directly due to drag, but it may also indirectly affect times by changing sprinters’ mindset in the race. Whereas drag created by headwinds and tailwinds work through the physics of air resistance, mindset effects work through their effect on the sprinters’ perception of speed and effort.

While there are clearly constraints on how fast an individual can run on a particular day, mindset often dictates whether the athlete reaches this full performance potential in a given race. The more seriously and intensively sprinters train, the stronger their mindset and the more consistently they will perform at close to biophysical constraints. Thus, when compared to the high school athletes, elite sprinters converge on the frontiers of what is possible given their

physical ability and the prevailing conditions for a given race.⁴

The direct effect of prevailing wind conditions on elite sprinter times in the 100 m is well-established (Linthorne 1994; Shanebrook and Jaszczak 1976). To isolate these direct physical effects of wind resistance on elite sprinters, we use the model proposed by Mureika (2001).⁵ Because the resulting wind-time function for elite sprinters is based only on the direct (projected) wind effect on performance through drag,⁶ it provides a clean benchmark for assessing the overall wind effects for high school sprinters, which we estimate statistically from race records.

We differentiate between direct wind effects and indirect mindset effects of wind using the drag equation parameterized for elite sprinters by Linthorne (1994) and a modified version (details in Section 5.1) for average boy and girl athletes. By comparing these effects, we demonstrate the psychological slack in high school sprinters’ performance, which creates the possibility of vicarious self-efficacy affecting sprint times.

2.2 Heterogeneity in beliefs & vicarious self-efficacy effects

The impact of vicarious experiences on an athlete’s self-efficacy depends, among other factors, on the salience of the experience and how comparable the other person’s performance is to oneself (Bandura 1977). Moreover, whether and how a self-efficacy effect translates into performance changes depends on the nature of the performance task and the scope for mindset to affect measured performance. Each of these observations raises potentially important heterogeneity dimensions in the context of a possible Boling effect.

Salience in this case requires that the other person—whose performance effectively models a level of success that was previously considered unattainable—is sufficiently comparable to oneself. The greater the similarities

⁴ Obviously, psychology can play a central role in elite sprinter performance as lapses in mental focus can lead individual sprinters to perform below their potential in a given race, but elite sprinters nonetheless serve as a convenient benchmark because their years of training allow them to approximate these biophysical limits.

⁵ The online calculator based on this model is available at <http://jmureika.lmu.build/track/wind/index.html> (Accessed 31 Aug 2021).

⁶ This function and the calculator on which it is based allows posted sprint times to be adjusted for wind effects in the absence of any (indirect) psychological effects. These model predictions match very closely the observed wind-time relationship for elite sprinters suggesting little or no measurable psychological effects among these sprinters.

one shares with the other person, the more potent the vicariousness of the other person's performance. In the behavioral economics domain, this is consistent with Appadurai's concept of an 'aspiration window' through which an individual determines whether others are similar enough to himself to be reliable models for what aspirations are reasonable to adopt (Appadurai 2004).

In the present context, one might therefore superficially conjecture that Boling's sprint would be most potent for white boys, particularly because the fact that Boling is white fueled the viral circulation of the performance on social media. It is important to note, however, that there is little empirical evidence about the subtleties of salience in these cases. It is certainly possible, for example, that Boling's sprint could have triggered a psychological reaction among all high school sprinters (i.e., if being a high school sprinter was sufficient basis for salience) or a different kind of belief response entirely among non-white sprinters. Furthermore, for a given athlete the general salience of race and ethnicity along with how this viral video was understood were surely shaped by the racial composition of his high school, team or competitors relative to himself. In our analysis, we treat these subtleties of salience heterogeneity as empirical questions and test for potential effects along many dimensions.

As for the mapping of enhanced self-efficacy into improved performance, the potentially asymmetric effects of headwinds and tailwinds during a race are a particularly useful feature of 100 m sprints. As described in the previous subsection, we use this feature to establish that psychological factors shape performance as a necessary condition for a Boling effect. To more fully appreciate how headwind and tailwind conditions alter the challenge of reaching one's potential as a sprinter, consider a few reflections on the likely source of these effects. Lining up in the starting blocks facing a stiff headwind can be discouraging. The sensation of added drag can be exasperating, even if it only lasts a few seconds. Such conditions test the focus and mental toughness of athletes. In contrast, tailwind conditions can create an exhilarating sensation of newfound speed, but even these conditions demand a certain kind of focus and mental toughness in order to continue exerting maximal effort when with less-than-full-effort one still feels fast. While basking in the sensation of speed, it is easy to back off just slightly.

These possible dimensions of heterogeneity correlate in potentially important ways with different U.S. states represented in our data.⁷ These state-level differences in

timing have two important implications for our analysis. First, since post-season meets get progressively more selective and competitive, the composition of sprinters represented in our data for a given state changes quickly with the onset of the post-season, which introduces a variety of complications for our DiD strategy. Second, these timing differences raise a temporal perspective on a potential effects on finishing times. The presence of slack in physical performance due to psychological factors opens the door for potentially immediate effects on performance due to enhanced self-efficacy beliefs (e.g., mental toughness in the face of a headwind). Beyond this immediate effect due to changes in beliefs—the Boling effect of interest—Boling's race might have also inspired more intense training and ultimately increased an athlete's physical potential as well as his mindset. Given these distinct state-level differences, we restrict our empirical test to meets in CA. In addition to accounting for more observations than any other state in our data, the CA schools in our data are racially diverse,⁸ and high school track meets in CA are much more likely to be equipped with the automated timing equipment needed to measure wind speed. More importantly, focusing on only CA holds constant the composition effects of post-season competition and enables us to isolate short-term mindset effects (i.e., the Boling effect) by closing down the training intensity pathway.

3 Data

We use data collected from a website used to post results from high school track and field meets from both 2018 and 2019.⁹ A unique athlete identifier tracks individual athletes both within and across these two years. Names of schools allows us to track high school track teams within and across years as well, in addition to enabling us to merge other school data (e.g., location and ethnic composition). In addition to athletes' times and grade level, we observe the name and date of the meet, whether the race was a preliminary

states are much more likely than others to have automated timing equipment that measures wind speeds during events. In states with relatively warm climates, the track season starts and ends early (e.g., Boling posted his record sprint in late April at the Texas high school championship meet), whereas the track season in colder states can stretch into late June two months later.

⁸ Moreover, diversity data for private schools are more readily available in CA than for other states.

⁹ There are a few such websites. We use the one that seems to get the most use among high schools, athletic.net. The majority of the high school teams represented in these data are based in the United States, but some Canadian high schools are represented as well.

⁷ In some states, track teams, meets and schools are quite ethnically homogeneous while others are very diverse. Track meets in some

Table 1: Summary statistics for 100 m sprints in 2018 and 2019 high school track meets recorded on athletic.net.

	Boys				Girls			
	All		CA only		All		CA only	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Time	12.603	(1.00)	12.677	(0.95)	14.557	(1.26)	14.647	(1.22)
2019	0.511	(0.50)	0.517	(0.50)	0.510	(0.50)	0.517	(0.50)
Post	0.298	(0.46)	0.126	(0.33)	0.308	(0.46)	0.132	(0.34)
Prelim heat	0.182	(0.39)	0.093	(0.29)	0.197	(0.40)	0.096	(0.30)
Finals	0.057	(0.23)	0.023	(0.15)	0.064	(0.25)	0.025	(0.16)
Varsity	0.768	(0.42)	0.509	(0.50)	0.787	(0.41)	0.535	(0.50)
JV	0.096	(0.29)	0.151	(0.36)	0.092	(0.29)	0.178	(0.38)
Automated times	0.170	(0.38)	0.432	(0.50)	0.177	(0.38)	0.441	(0.50)
Race size	33.205	(26.58)	29.990	(34.15)	29.169	(23.32)	27.694	(28.08)
Time gap	0.126	(0.21)	0.140	(0.22)	0.171	(0.26)	0.184	(0.28)
N	708,706		116,339		612,004		99,148	

'Time' is recorded finishing time in seconds. '2019' indicates races run in 2019. 'Post' indicates races held during the post-period in both 2018 and 2019. 'Prelim Heat' indicates races that were held to qualify for the final 100 m race of the meet. 'Finals' indicates races that were part of the final 100 m for a given meet. 'Varsity' indicates that the race was at the varsity level (i.e., most advanced in high school sports). 'JV' indicates that the race was at the junior varsity level. 'Automated times' indicates that the meet used automated timing equipment. 'Race size' indicates the total number of 100 m sprinters in a given meet. 'Time gap' indicates the time in seconds between a given sprinter and the next closest finisher behind him or her.

or final heat, time gaps between runners for a given race, whether the race was at the varsity or junior varsity (JV) level, the total number of competitors in a particular event at a given meet, and whether the meet was timed using 'fully automated timing' equipment. For meets with automated timing equipment, which tend to be more competitive on average, we also observe the wind speed registered at the time of the 100 m race.¹⁰

The race results recorded online that are the basis for this analysis are quite clean. There are, however, some clear errors in the data, which we clean by dropping any times that are faster than current records. We also trim the slowest 1 % of times for boys and girls 100 m races. We exclude wheelchair and para-ambulatory races, which constitute a tiny fraction of the total observations. While we use athletes across all states for some of the analysis that follows, as explained in the preceding section we restrict our focus to those competing in CA when quantifying the potential impacts of Boling's race on high school sprinter times.

Table 1 shows descriptive statistics for our outcome variable of interest (Time) and a set of other variables that

feature in our subsequent specifications. For both boys and girls 100 m races, we show these summary statistics for all states and for CA only. The average 100 m time for boys is about 2 s faster than for girls. Total observations for boys and girls (and for all states and CA only) are split evenly between 2018 and 2019 (51 % in 2019). The indicator variable 'Post' defines the post-period for the DiD empirical strategy described below. Specifically, in 2019 we set Post = 1 after the April 27, 2019 meet in Texas at which Boling set the all conditions 100 m record. In 2018, this same meet was held on April 28, 2018, but the post-season competition in CA concluded with the state championship meet one week later that year. To reflect this shift in the timing of the competitive and selective post-season meets, we define the post-period for this comparison year by setting Post = 1 after May 5, 2018.¹¹ About 30 % of all observations across these events come from meets held in this later portion of the season, which we will refer to as the post-period. In CA, the share of post-period observations drops to about 13 % because the track and field season in CA starts and ends earlier than many states in colder northern locations. In CA, the

¹⁰ Standard track rules dictate that the anemometer used to gauge wind speed be placed 50 m from finish line within 2 m of the inside lane and aligned (i.e., parallel) to the track. The gauge is often encased in a plastic tube so that only direct headwinds, tailwinds, or the effective parallel vector of crosswinds is detected. The gauge is started with the starting gun of the 100 m race, and the average measured wind speed for the subsequent 10 s is recorded as the official wind for a given race.

¹¹ While all of the observations from 2018 of course precede Boling's 100 m feat, this 2018 post-period serves as something of a 'control' for the 2019 post-period during which sprinters are effectively 'treated' by having seen his sprint. By defining the post-period relative to the start and end of the track season in CA, we ensure that the DiD comparison uses comparable weeks of competition relative to the championship meet. This is obviously far more relevant for track athletes in CA than the date on which the 2018 Texas state championship occurred.

post-period coincides closely with post-season competitions that become progressively more competitive and culminate in the state championship meet held in late May. Whereas 17–18 % of all meets in our data were equipped with automated timing equipment, this is much higher (43–44 %) in CA.

As a preliminary look at the times we use in the analysis below, we show in Figure 1 the full distribution of 100 m

times in CA for boys and girls in 2018 and 2019. Foreshadowing our DiD approach described in the next section, we display these distributions separately by pre- and post-periods (i.e., according to ‘Post’ in Table 1) for each year. The ‘2018 pre’ and ‘2019 pre’ distributions of finishing times are indistinguishable from each other. The post-period distributions are completely to the left of the pre-period distributions, which is to be expected because (i) athletes tend to get faster

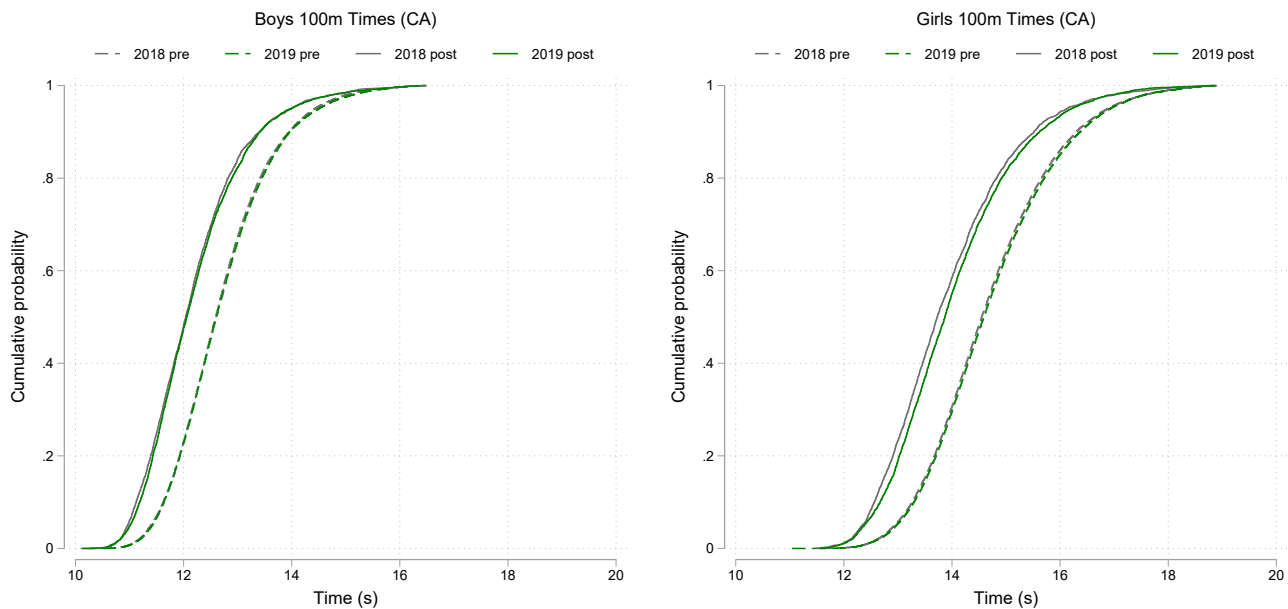


Figure 1: Cumulative distributions of times for boys and girls 100 m races in CA. Distributions shown separately for pre- and post-periods in 2018 and 2019 as defined for the purposes of DiD specification.

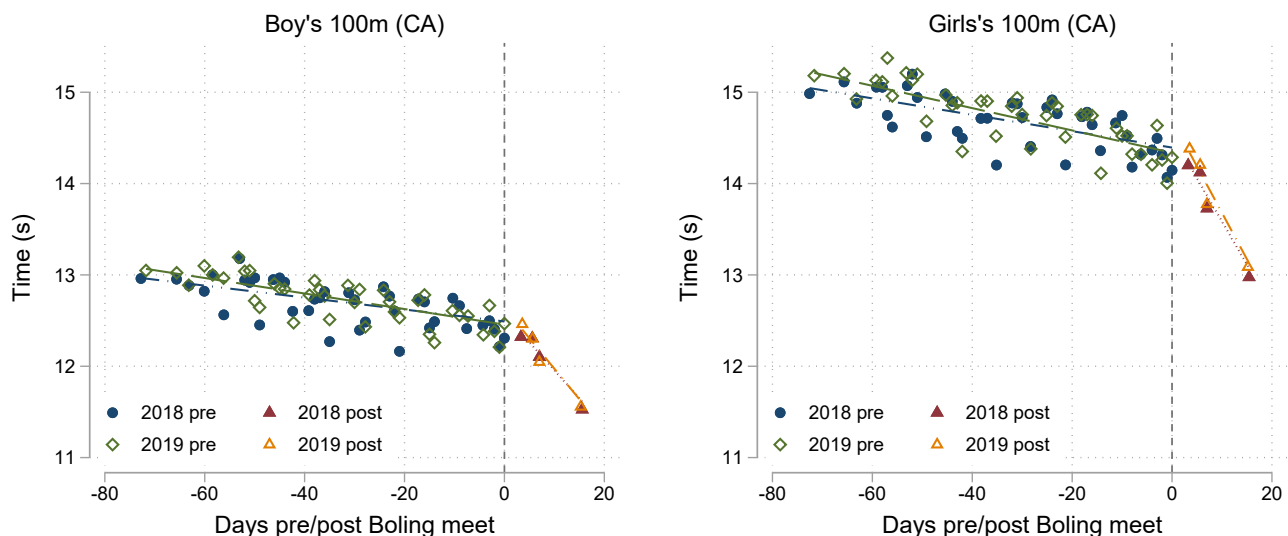


Figure 2: Non-parametric visualization of times for boys and girls 100 m races in CA using a binned scatterplot (each point represents the average time of 50 equally-sized bins by date of meet). The post-period for 2019 is defined by the actual date of Boling’s record-breaking sprint (i.e., the ‘Boling meet’). The post-period for 2018 is defined by the same point relative to the start and end of the CA high school track season and constitutes a ‘pseudo’ post-period as a DiD comparison with the true post-period in 2019.

as the season progresses and, more importantly, (ii) the post-period coincides in CA with the start of more competitive and selective post-season meets for which only fast sprinters qualify. For girls a slim wedge separates the 2018 and 2019 post-period distributions with faster times recorded in the 2018 post-period. For boys, post-period distributions are nearly indistinguishable.

Progressive improvement across the track and field season is a key feature of our data, as is the onset of the increasingly selective post-season that ends with the state championship meet. To visually depict these aspects, Figure 2 shows binned scatterplots of finishing times across the season—relative to the Boling meet—for boys and girls.¹² While pre-period average times are consistent with gradual improvement due to training and experience, the selective post-period shows more dramatically faster times as the fastest athletes progress through post-season meets.¹³ Again, the fact that the average times and trend prior to the Boling meet are nearly indistinguishable in 2018 and 2019 bodes well for our DiD identification strategy.¹⁴

As a methodological innovation, we modify the algorithm published in Voicu (2018) to predict the race of each athlete in our data.¹⁵ Predicted sprinter race and racial composition of teams enable us to test for a wider range of relevant heterogeneous Boling effects. Specifically, we use the BIFSG predicted probabilities that a given boy is white and average $\Pr(\text{white})$ across high school teams (see Appendix). We modify the BIFSG algorithm for this application by integrating school-level racial composition data as the basis for the location predictions of race, instead of relying on census block data, which can only improve the resulting racial predictions.¹⁶

¹² These scatterplots depict average times in 50 equally-sized bins across the 2018 and 2019 CA track and field season.

¹³ Every new season replicates virtually the same pattern as previous seasons because the composition of high school sprinters remains statistically unchanged from one season to the next: Every year, new freshman athletes begin to compete with returning sprinters who hope to build on their prior experience and on being one year older.

¹⁴ It is important to remember, however, that these unconditional averages do not control for other factors that affect sprint times and therefore do not represent a clean test of a Boling effect.

¹⁵ This algorithmic approach to predicting race based on names has been extended more recently to account for common problems with census data using a ‘fully’ Bayesian approach (Imai et al. 2022).

¹⁶ These school-level racial composition data are reported annually by all public schools and available at <https://nces.ed.gov/ccd/schoolsearch/>. In California, many private schools also report these statistics, which are available at <https://www.cde.ca.gov/ds/>. When we have racial composition at the school-level from these sources, we use this as the basis for location-based predictions of race. For those

4 Empirical approach

We describe in this section the two main empirical analyses that follow. First, we present the specification we use to estimate the effect of wind on 100 m sprint times. Then, we discuss the DiD approach we use to test for the presence of a Boling effect and the triple DiD (3DiD) approach we use to test for heterogeneous effects by (predicted) race of athletes.

In order to estimate how wind affects high school sprint times and to test for differences in these wind effects by gender, we use the following piecewise-linear specification, which we estimate separately for boys and girls in 100 m sprint races:

$$\text{Time}_{ijt} = \alpha_0 + \alpha_1 W_{jt} + \alpha_2 [W_{jt} \times \text{Head}_{jt}] + X'_{ijt} \theta + \varepsilon_{ijt} \quad (1)$$

where Time_{ijt} is the time posted by athlete i in race j on date t , W_{jt} is the wind as recorded during the race, Head_{jt} is an indicator for races conducted in headwind conditions (i.e., $W_{jt} < 0$), and X_{ijt} is a vector of control variables including the time gap at the finish line to the next fastest finisher, the date, type and size of the race, and fixed effects for athletes’ grade (i.e., year in high school (e.g., freshman, sophomore, etc.)).¹⁷ We report results from this estimation with and without athlete fixed effects. In this specification, $\alpha_1 < 0$ indicates that stronger tailwinds (headwinds) decrease (increase) finishing times, and $\alpha_2 < 0$ indicates the additional time effect of a headwind. By comparing these wind effects for boys and girls with the wind effects for elite sprinters, we can infer the presence of psychological factors in high school sprint performance. In other words, we posit that the systematic effect of a headwind on finishing times over and above its impact via wind resistance is attributable to psychological effects of the wind conditions that prevail at the time of a race.

We estimate Equation (1) using all races with automated equipment to record wind speed conducted in 2018 and 2019,

private schools for which such data is not publicly available, we use the racial composition of the zip code of the school as the basis for these location predictions. Using racial composition at the school-level has two main advantages: (i) high schools generally include fewer students than census blocks include people, implying that school-level data are typically higher resolution spatially than census block data, and (ii) these school-level racial composition data are updated every year rather than every decade, implying that these data are higher resolution temporally than census data.

¹⁷ We attempted to use lasso to select these control variables for this specification and the DiD and 3DiD specifications below, but discovered that all available control variables, which we had already included, were retained.

except those held after Boling's race on April 27, 2019.¹⁸ As an alternative specification, we estimate a panel version of Equation (1) that includes athlete fixed effects. We also estimate a version of this panel specification that includes quadratic wind speed to allow for more flexible wind effects on finishing times.

As part two of our empirical analysis, we use a DiD specification to test for possible impacts on subsequent race times. Attributing differences in sprint times after Boling's race to a "Bolling effect" requires a causal identification strategy. In our case, no such strategy will be perfect given our use of observational rather than experimental data. We use 2018 as the counterfactual for 2019 based on the assumption that these two years are comparable up until Boling's 9.98 s race marked the beginning of the post-period in 2019. The indistinguishable pre-period distributions in Figure 1 provide initial visual evidence that these two years are comparable in this way.

This identification strategy also assumes that the timing of Boling's record-breaking race was random. This is supported by the fact that his time was aided by a tailwind of +4.2 m per second (m/s) that was strong enough to push his time below the symbolic threshold of 10 s and created the buzz that caused his performance to go viral. To underscore this point, two points are noteworthy. First, if the wind at the time of the race had not gusted to such a stiff tailwind, Boling almost certainly would not have broken the 10 s barrier, and the sprint would not have gone viral. The boys 100 m final was the only wind-registered race held at that meet that day with wind recorded over ± 2 m/s.¹⁹ This random (and essential) wind gust during the boys 100 m final rules out the presence of other date-specific factors affecting both Boling's performance and other high school sprinters' subsequent times. Second, since that record-setting sprint, Boling has only broken the 10 s barrier twice at official events: one of those was also wind assisted (+3.2 m/s) and the other was within the allowable tailwind range (+1.6 m/s). The confluence of factors that enable a talented athlete like Boling to run 100 m under 10 s are simply very rare and entail an element of randomness in race conditions.

Conditional on this DiD strategy producing at least weakly causal estimates, we hypothesize that any changes

in 100 m times are attributable to mindset changes since, as mentioned, the post-period observations in CA occur too soon after Boling's race for increased training intensity to increase athletes' *physical* potential. While we believe this is a defensible approach to estimating the Boling effect, it clearly does not approximate an ideal (hypothetical) experiment that randomly varies exposure to Boling's race and does not entirely rule out possible confounding factors. In the absence of such an experiment, our second-best DiD approach (further enhanced with additional adjustments described below) provides suggestive evidence of an effect that is unlikely to arise purely from correlation.

Our basic DiD specification is as follows:

$$\text{Time}_{ijt} = \beta_0 + \beta_1 2019_{jt} + \beta_2 \text{Post}_{jt} + \beta_3 [2019_{jt} \times \text{Post}_{jt}] + X'_{ijt} \theta + \varepsilon_{ijt} \quad (2)$$

where Time_{ijt} is the time posted by athlete i in race j on date t , β_3 is the DiD coefficient of interest that indicates the Boling effect on subsequent times, and X_{ijt} is a vector of control variables including the time gap at the finish line to the next fastest finisher, the date, type and size of the race, and whether the meet was using fully automated timing equipment.²⁰ We restrict our focus to CA schools and meets as described above and estimate this DiD specification separately for boys and girls. In addition to the specification in (2), which is estimated using all meets in our data and controlling for those with automated timing, we also estimate a version of this specification separately for races run in headwind and tailwind conditions (and therefore only including only races with automatic timing).

Evidence of Boling's performance inspiring faster times would come in the form of $\beta_3 < 0$. As described in the conceptual framework above, there are, however, a host of potentially important heterogeneity considerations to take into account. While boys may be most responsive to the Boling effect since he posted his record time in the boys 100 m, the scope of salience in this case could easily extend to girls competing in the 100 m sprints. To further explore this scope of salience in this case, we include racial predictions in two ways. First, we estimate Equation (2) separately by tercile of team average $\text{Pr}(\text{white})$ (maintaining the disaggregation by gender and wind conditions). Second, we estimate a 3DiD

¹⁸ We exclude races held after Boling's record setting performance to avoid any potential confounding effects of this race, which we directly test for in the DiD specification described below. We include data from races in other U.S. states, but the majority of the observations come from CA meets.

¹⁹ Wind-registered races held that day include the boys and girls preliminary heats and finals for both the 100 m and 200 m events.

²⁰ Because the sample of sprinters shrinks as the post-season progresses, we do not include athlete fixed effects in this specification.

specification as follows:

$$\begin{aligned} \text{Time}_{ijt} = & \beta'_0 + \beta'_1 2019_{jt} + \beta'_2 \text{Post}_{jt} + \beta'_3 [2019_{jt} \times \text{Post}_{jt}] \\ & + \gamma_1 \text{Pr}(\text{white}) + \gamma_2 [2019_{jt} \times \text{Pr}(\text{white})] \\ & + \gamma_3 [\text{Post}_{jt} \times \text{Pr}(\text{white})] \\ & + \gamma_4 [2019_{jt} \times \text{Post}_{jt} \times \text{Pr}(\text{white})] + X'_{ijt} \theta + \varepsilon_{ijt} \end{aligned} \quad (3)$$

where β'_3 is the base DiD coefficient and γ_4 is the 3DiD coefficient that provides a further adjustment of finishing times based on the predicted $\text{Pr}(\text{white})$ of athlete i . As with the DiD specification, we separately estimate this equation by gender, wind conditions and team average $\text{Pr}(\text{white})$. Evidence that the vicarious self-efficacy Boling effect is concentrated in white sprinters would come with the result $\gamma_4 < 0$.

We make two statistical adjustments when estimating these DiD and 3DiD specifications to improve identification and statistical inference. First, to strengthen our DiD identification strategy, which takes athletes competing in 2018 as our counterfactual for those competing in 2019, we use entropy balancing (Hainmueller 2012; Hainmueller and Xu 2013) to ensure balance between these “control” and “treated” athletes. We then use the resulting entropy balancing weights (EBW) in our DiD and 3DiD regressions. This adjustment is potentially important because while the composition of high school sprinters between the post-period of 2018 and 2019 in CA is surely comparable, the composition of those included in the online results platform we use may not be.²¹ Entropy balancing and weighted regression using EBWs ensures that the pre-period means of key covariates in 2018 match their counterparts from 2019.²²

²¹ From one year to the next, schools or entire districts or leagues can shift from one online results platform to another, which could alter the composition of athletes. Using EBW alleviates any concern that such changes in composition confound our DiD estimation.

²² We match the following covariates for all athletes as well as for the sub-set of athletes who compete in the post-period: Average time, standard deviation of times, proportion of races run in races with automatic times, and the season best time for these athletes. As a robustness check, we also tried using inverse probability weights to balance “control” and “treated” athletes. The set of results for the DiD and 3DiD specifications are qualitatively unchanged, so we report only the EBW results. For interested readers, we describe this alternative weighting approach here. We construct inverse-probability weights using a first-stage logit lasso model of treatment status. We use ten-fold cross validation to tune the parameters and 15 variables were selected by the model, including the number of races run (and squared), average time (and squared), standard deviation of times, grade, proportion of races run in races with automatic times (and squared), proportion of races run at the varsity level

As the second adjustment, we correct for multiple hypothesis testing since we separately test heterogeneity by gender, wind conditions, and racial composition of teams. In order to reduce the likelihood of false rejections when we separately estimate the same DiD (or 3DiD) specification for several sub-samples of our data, we compute sharpened False Discovery Rate (FDR) q -values (Anderson 2008), which we report in the results tables along with standard errors. The sharpened q -values are the results of a two-stage Benjamini–Hochberg (BH) procedures that incorporates the number of hypotheses rejected in the first stage into the second stage to achieve greater power.

5 Results

In this section, we present and discuss results from our two sets of analyses. First, we estimate the effect of headwinds and tailwinds on high school sprinter performance. These results establish the prevalence of systematic indirect effects of wind on finishing times that work through psychological channels. We contrast these wind effects for the average high school sprinter to those of elite sprinters and differentiate these effects for boys and girls. Second, we estimate DiD and 3DiD specifications to test for a Boling effect using high school track meets in CA. These results, which should be interpreted with the causality caveats discussed above, reveal a number of potentially important dimensions of heterogeneity in this effect, including by gender and predicted racial composition of one's team.

5.1 Gender-differentiated wind effects on performance

Table 2 reports the estimation results from Equation (1) with and without athlete fixed effects (FE). While we focus primarily on the two coefficients on Wind, the control variables are generally statistically significant and contribute to the relatively high R^2 . The FE results suggest that the tailwinds reduce finishing times equally for boys and girls, but that headwinds penalize girls' performance more than boys. To see this asymmetric gender difference graphically, we estimate a version of this FE specification separately for

and a set of dummies for these variables indicating missing values. The propensity scores are then predicted at the athlete-level and then standardized to be merged back to their corresponding race records.

Table 2: Wind effect on 100 m times for boys and girls for high school track meets with automated timing equipment with and without athlete fixed effects (FE).

	Boys		Girls	
	Without athlete FE	With athlete FE	Without athlete FE	With athlete FE
Wind	−0.055*** (0.002)	−0.050*** (0.001)	−0.064*** (0.003)	−0.049*** (0.001)
Headwind × wind	−0.042*** (0.004)	−0.037*** (0.002)	−0.061*** (0.006)	−0.053*** (0.002)
Time gap	2.125*** (0.013)	0.562*** (0.008)	2.150*** (0.017)	0.345*** (0.008)
Finals	−0.712*** (0.023)	−0.021* (0.011)	−0.997*** (0.033)	−0.039*** (0.012)
Prelim heat	−0.145*** (0.010)	0.069*** (0.005)	−0.175*** (0.015)	0.078*** (0.006)
Varsity	−0.104*** (0.005)	−0.002 (0.003)	−0.047*** (0.008)	−0.010*** (0.004)
JV	0.351*** (0.010)	−0.004 (0.005)	0.643*** (0.014)	−0.023*** (0.006)
Race size	0.008*** (0.000)	0.001*** (0.000)	0.014*** (0.000)	0.001*** (0.000)
Race size ²	−0.000*** (0.000)	−0.000*** (0.000)	−0.000*** (0.000)	−0.000*** (0.000)
Date	−0.004*** (0.000)	−0.005*** (0.000)	−0.007*** (0.000)	−0.007*** (0.000)
Finals × size	0.023*** (0.002)	−0.000 (0.001)	0.033*** (0.003)	0.002* (0.001)
Prelim × size	−0.004*** (0.000)	−0.001*** (0.000)	−0.006*** (0.000)	−0.001*** (0.000)
Finals × time gap	−0.763*** (0.059)	0.277*** (0.025)	−0.988*** (0.077)	0.249*** (0.026)
Constant	12.693*** (0.011)	13.024*** (0.007)	14.356*** (0.016)	14.828*** (0.008)
R-squared	0.44	0.40	0.35	0.35
Observations	102,920	102,920	90,510	90,510

Data from U.S. high school track meets in 2018 and 2019 that used automated timing equipment (including wind speed and direction) and were posted at athletic.net. Standard errors in parentheses. Grade fixed effects included. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

headwind and tailwind conditions, allowing for non-linear (quadratic) wind effects and show the marginal effects of wind in Figure 3. The headwind penalty is particularly pronounced for girls: the same 6 m/s headwind slows boys by 3.2 % and girls by 3.8 % on average.

As described above, to test whether these wind effects for high school sprinters include a combination of physics and psychology we next compare these effects to elite male sprinters. If this test suggests the presence of measureable mindset effects and therefore slack in high school sprint performance relative to hard biophysical constraints, it seems plausible that we might find evidence of a Boling effect on performance in the subsequent DiD analysis. The left panel of Figure 4 overlays the wind-time function for elite

sprinters based on Mureika's online calculator²³ on the linear wind-time functions implied by the FE estimates for high school boy and girl sprinters in Table 1. While faster times due to tailwinds for boys and girls are nearly identical to the benchmark wind-time function of elite sprinters, boys and (especially) girls suffer much slower times due to headwinds.

The right panel of Figure 4 depicts these functions with the added dimension of the drag produced by headwinds and tailwinds. For elite sprinters, we use the drag equation and parameterization of Linthorne (1994). We adapt this

²³ See <http://jmureika.lmu.build/track/wind/index.html> and see discussion above.

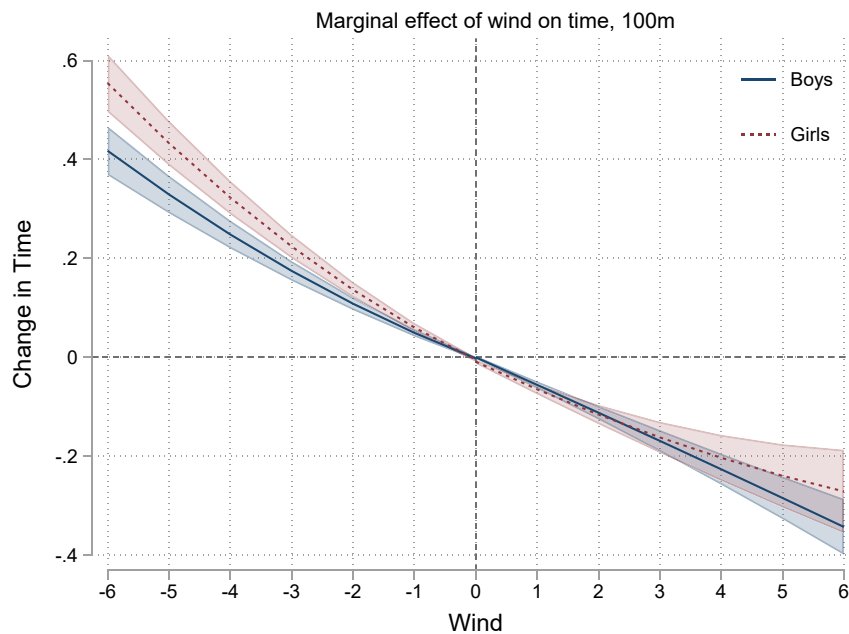


Figure 3: Quadratic effect of wind (m/s) on 100 m times (s) for high school boys and girls based on specification with athlete fixed effects. (Wind < 0 indicates headwind).

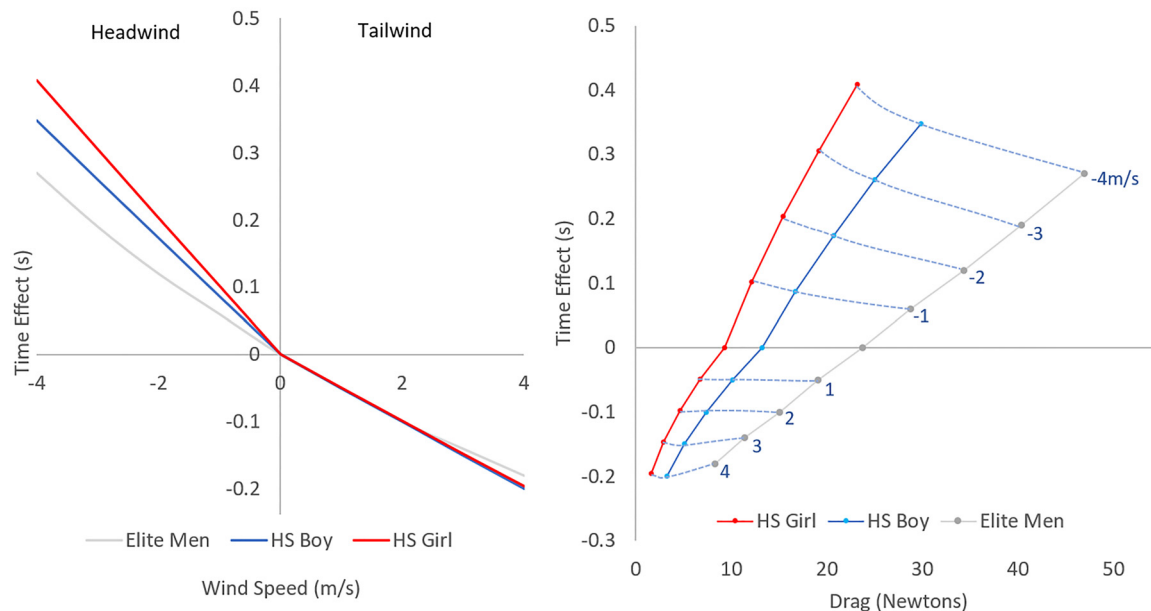


Figure 4: Estimated linear wind effects for HS boys and girls compared to elite male sprinters in 100 m races (left) and wind effects as a function of drag created by prevailing winds during the race (right).

equation for high school sprinters using the average times (velocity) in our data and modified drag areas for boys and girls, respectively. Elite sprinters obviously generate greater drag in any wind condition than high school sprinters due to the combination of their greater speed and (generally) larger stature. Visually adjusting for this difference in drag accentuates the asymmetric effect created by headwinds, which is evident in the kink that occurs in

the transition from tailwinds to headwinds as well as in the sloping (dashed) iso-wind lines that connect the functions in the right panel. Based on this evidence of physical slack in sprinter performance due to psychological factors, it seems conceivable that Boling's record setting sprint could have affected performance—even in the very short-run—by vicariously boosting the self-efficacy of high school sprinters.

Table 3: DiD results for the 100 m event in high school track meets.

	Boys 100 m			Girls 100 m		
	All	Tailwind	Headwind	All	Tailwind	Headwind
2019	0.015*** (0.005)	−0.022** (0.009)	0.049*** (0.011)	0.017*** (0.006)	0.005 (0.012)	−0.004 (0.015)
Post	−0.048*** (0.014)	−0.197*** (0.023)	−0.068** (0.029)	−0.187*** (0.019)	−0.437*** (0.031)	−0.273*** (0.040)
2019 × post	−0.010 (0.017) [1.000]	0.183*** (0.028) [0.001]	−0.009 (0.037) [1.000]	0.042* (0.023) [0.024]	0.183*** (0.038) [0.001]	0.119** (0.048) [0.015]
Constant	12.502*** (0.010)	11.519*** (0.021)	11.584*** (0.027)	14.307*** (0.015)	13.772*** (0.029)	13.643*** (0.038)
R-squared	0.40	0.43	0.40	0.46	0.47	0.47
Observations	106,550	28,880	17,304	84,267	23,755	14,251
Average time	12.68	12.40	12.68	14.65	14.29	14.61

Data taken from 2018 to 2019 meets in CA. Estimated DiD coefficient (i.e., Boling effect) corresponds to variable 2019 × Post. Observations from 2018 are weighted by entropy balancing to match mean 2019 values of the following covariates for all athletes as well as for those competing in the post-period: average time, standard deviation of time, proportion of races with automatic timing, and the fastest time of the athletes. Included but not reported are grade fixed effects, other control variables from Table 2, and the number of days remaining in the season. Standard errors in parentheses. Sharpened q -values for estimated DiD coefficients reported in square brackets. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5.2 Vicarious self-efficacy & sprinter performance

In this section, we first estimate the simple DiD specification in Equation (2) and show these results in Table 3. A variety of differences between boys and girls are clear in these results. Throughout, we have roughly three times more observations for boys than girls, which tends to increase precision for estimated coefficients for boys. The coefficient on Post is uniformly negative (although larger in magnitude for girls), which is expected since this indicator variable coincides with the post-season in high school track and field in CA.

Since our focus is primarily on the estimated Boling effect as captured by the DiD coefficient on 2019 × Post (β_3), we provide sharpened q -values in addition to standard errors for these estimates.²⁴ For boys, the ‘All’ DiD coefficient is negative, but statistically insignificant. For girls, this coefficient is positive and statistically significant, suggesting a mindset effect that *slowed down* rather than speeding up sprinters. When we restrict our sample to the automated races that record wind speed and direction and estimate separate tailwind and headwind regressions, we see that boys also experience a *flipped* Boling effect in tailwind conditions. Coincidentally, the tailwinds point estimate for girls

is identical to the boys’. In mentally taxing headwind conditions, we see no effect for boys, but continue to see a flipped effect for girls. The headwinds-tailwinds asymmetry for boys is puzzling as it seems to suggest that instead of instilling greater mental toughness in challenging headwind conditions that normally produce slack in performance (see Figure 4), the Boling effect rather *introduced* slack in their performance under more favorable tailwind conditions. This flipped effect, which in tailwind races decreases performance by 1.5 % and 1.3 % for boys and girls, respectively, may be due to overconfidence mid-race when—aided by the wind and feeling fast like Boling—sprinters let up just slightly (i.e., if a race feels effortless, it may induce less effort). To be clear, such overconfidence is ever-present in tailwind conditions. The estimates suggest that Boling’s race may have exacerbated this overconfidence by affecting the beliefs sprinters bring to the starting blocks. Whatever the explanation, the presence of this flipped effect suggests a composite of opposing vicarious forces rather than a simple uniform effect.

Next, we unpack these DiD results by separately estimating these regressions by team average Pr(White) terciles (see Figure A1). By allowing the estimated effect to differ based on the (predicted) racial composition of each athlete’s sprint team,²⁵ we reflect the fact that the viral video of

²⁴ These q -values correct for multiple hypothesis testing across the three separate regressions (All, Tailwind, Headwind) for boys and girls.

²⁵ Average Pr(white) is computed separately by boys and girls for a given school and is based on the predicted Pr(white) for all boys (or girls) from a given school who compete in at least one 100 m sprint.

Table 4: DiD results for boys by team average Pr(white) terciles.

	All			Tailwind			Headwind		
	Low	Mid	High	Low	Mid	High	Low	Mid	High
2019	0.021*** (0.008)	0.027*** (0.008)	−0.009 (0.008)	−0.027* (0.015)	0.000 (0.014)	−0.053*** (0.015)	0.042** (0.019)	0.115*** (0.020)	0.024 (0.021)
Post	−0.018 (0.025)	−0.066*** (0.025)	−0.043* (0.023)	−0.297*** (0.044)	−0.195*** (0.040)	−0.135*** (0.037)	−0.026 (0.052)	−0.088* (0.049)	0.106** (0.053)
2019 × post	−0.019 [0.416]	−0.052* [0.161]	0.035 [0.284]	0.296*** [0.001]	0.108** [0.059]	0.220*** [0.001]	−0.059 [0.321]	−0.047 [0.416]	−0.065 [0.321]
Constant	12.509*** (0.017)	11.797*** (0.019)	12.543*** (0.017)	12.149*** (0.032)	11.996*** (0.029)	11.367*** (0.034)	11.764*** (0.043)	12.363*** (0.041)	12.470*** (0.041)
R-squared	0.39	0.40	0.41	0.41	0.43	0.46	0.40	0.39	0.40
Observations	35,009	35,875	35,339	8,358	10,497	9958	5553	5965	5733

Data taken from 2018 to 2019 meets in CA. Estimated DiD coefficient (i.e., Boling effect) corresponds to variable 2019 × Post. Observations from 2018 are weighted by entropy balancing to match mean 2019 values of the following covariates for all athletes as well as for those competing in the post-period: average time, standard deviation of time, proportion of races with automatic timing, and the fastest time of the athletes. Included but not reported are grade fixed effects, other control variables from Table 2, and the number of days remaining in the season. Terciles are based on ‘average by team’ predicted Pr(white) (see Figure A1). Standard errors in parentheses. Sharpened q -values for estimated DiD coefficients reported in square brackets. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Boling’s sprint frequently accompanied commentary about his race. Table 4 shows the DiD results for boys disaggregated by these Pr(white) terciles. In the ‘All’ races results, the middle tercile (i.e., athletes competing on sprint teams that are most racially mixed) has an effect that is different than the high and low terciles (i.e., athletes competing on sprint teams that are predominantly white or non-white). For these middle tercile sprinters, the effect is consistent

with vicarious self-efficacy as they post faster times after Boling’s sprint went viral. Although this Boling effect is statistically significant by conventional standards, this significance fades based on the q -values that correct for multiple testing. For tailwind races, we see a flipped effect across all three terciles (albeit smaller in the middle tercile). All terciles have negative point estimates in headwind conditions, but none are statistically significant. For girls (Table 5),

Table 5: DiD results for girls by team average Pr(white) terciles.

	All			Tailwind			Headwind		
	Low	Mid	High	Low	Mid	High	Low	Mid	High
2019	0.034*** (0.011)	0.027** (0.011)	−0.043*** (0.011)	−0.003 (0.022)	0.066*** (0.020)	−0.068*** (0.021)	0.029 (0.027)	−0.010 (0.026)	−0.089*** (0.028)
Post	−0.192*** (0.034)	−0.173*** (0.033)	−0.165*** (0.030)	−0.658*** (0.063)	−0.395*** (0.056)	−0.353*** (0.049)	−0.460*** (0.073)	−0.223*** (0.070)	−0.030 (0.071)
2019 × post	0.030 [0.672]	0.017 [0.983]	0.090** [0.025]	0.297*** [0.001]	0.072 [0.428]	0.267*** [0.001]	0.291*** [0.002]	0.012 [1.000]	0.005 [1.000]
Constant	14.435*** (0.027)	14.211*** (0.026)	14.150*** (0.027)	13.715*** (0.053)	13.278*** (0.048)	13.440*** (0.048)	14.347*** (0.057)	13.863*** (0.059)	13.852*** (0.063)
R-squared	0.46	0.47	0.46	0.47	0.48	0.46	0.46	0.47	0.46
Observations	26,910	27,432	27,272	7032	8113	7856	4332	4930	4449

Data taken from 2018 to 2019 meets in CA. Estimated DiD coefficient (i.e., Boling effect) corresponds to variable 2019 × Post. Observations from 2018 are weighted by entropy balancing to match mean 2019 values of the following covariates for all athletes as well as for those competing in the post-period: average time, standard deviation of time, proportion of races with automatic timing, and the fastest time of the athletes. Included but not reported are grade fixed effects, other control variables from Table 2, and the number of days remaining in the season. Terciles are based on ‘average by team’ predicted Pr(white) (see Figure A1). Standard errors in parentheses. Sharpened q -values for estimated DiD coefficients reported in square brackets. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 6: 3DiD results for boys 100 m by team average Pr(white) terciles.

	All			Tailwind			Headwind		
	Low	Mid	High	Low	Mid	High	Low	Mid	High
2019	0.029*** (0.009)	0.026** (0.012)	−0.036** (0.018)	−0.016 (0.018)	0.013 (0.022)	−0.063** (0.031)	0.035 (0.022)	0.125*** (0.030)	0.020 (0.045)
Post	−0.013 (0.029)	−0.066* (0.037)	−0.130** (0.051)	−0.280*** (0.053)	−0.180*** (0.064)	−0.222*** (0.082)	−0.025 (0.062)	−0.139* (0.071)	−0.117 (0.107)
2019 × post	−0.015 (0.037)	−0.124** (0.049)	0.159** (0.068)	0.289*** (0.064)	0.049 (0.078)	0.349*** (0.105)	−0.041 (0.075)	−0.125 (0.104)	0.234 (0.150)
Pr(white)	−0.291*** (0.024)	−0.135*** (0.016)	−0.189*** (0.017)	−0.244*** (0.045)	−0.104*** (0.028)	−0.200*** (0.030)	−0.282*** (0.053)	−0.142*** (0.041)	−0.229*** (0.042)
2019 × Pr(white)	−0.083** (0.034)	−0.017 (0.023)	0.053** (0.024)	−0.107 (0.065)	−0.055 (0.040)	0.023 (0.042)	−0.045 (0.077)	−0.082 (0.055)	0.018 (0.060)
Post × Pr(white)	−0.079 (0.104)	0.003 (0.069)	0.139** (0.065)	−0.108 (0.185)	−0.021 (0.112)	0.148 (0.104)	−0.061 (0.215)	0.101 (0.130)	0.352*** (0.136)
2019 × post × Pr(white)	0.023 [0.693]	0.187** [0.144]	−0.180** [0.144]	0.076 [0.693]	0.123 [0.517]	−0.182 [0.294]	−0.028 [0.693]	0.254 [0.294]	−0.484** [0.144]
Constant	11.964*** (0.020)	11.869*** (0.021)	11.915*** (0.023)	11.684*** (0.036)	11.349*** (0.035)	11.508*** (0.041)	11.803*** (0.046)	11.748*** (0.051)	11.759*** (0.059)
R-squared	0.40	0.41	0.42	0.42	0.44	0.47	0.41	0.39	0.41
Observations	32,348	32,544	31,390	7679	9537	8760	5071	5474	5061

Data taken from 2018 to 2019 meets in CA. Estimated 3DiD coefficient corresponds to variable 2019 × Post × Pr(white). Observations from 2018 are weighted by entropy balancing to match mean 2019 values of the following covariates for all athletes as well as for those competing in the post-period: average time, standard deviation of time, proportion of races with automatic timing, and the fastest time of the athletes. Included but not reported are grade fixed effects, other control variables from Table 2, and the number of days remaining in the season. Terciles are based on ‘average by team’ predicted Pr(white) (see Figure A1). Standard errors in parentheses. Sharpened q -values for estimated DiD coefficients reported in square brackets. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

DiD coefficients across all race conditions and terciles are positive, but with mixed statistical precision. In tailwind conditions, girls on racially-mixed teams show no effect, but those in low and high tercile teams seem to slacken their pace after Boling’s performance.

Finally, we turn to the 3DiD specification in Equation (3). The results for boys from this 3DiD regression are shown in Table 6. We continue to see evidence of the Boling effect concentrated in the racially-mixed middle tercile. Indeed, this effect is both larger in magnitude and more precisely estimated and seems to be driven more by headwind than tailwind conditions. The estimated 3DiD coefficients reveal how these vicarious effects appear to vary by both the race of the sprinter and the racial composition of his team. More specifically, ‘All’ race results hint²⁶ that the Boling effect on racially mixed teams may be concentrated in non-white

athletes (i.e., with low Pr(white)).²⁷ No clear effect is evident for white athletes on these mixed teams (given by the statistically zero sum of the DiD and 3DiD coefficients). ‘All’ results are different for predominantly white, high tercile teams: Their non-white athletes are *slower* in the wake of Boling’s sprint. The Boling effect disparity between white and non-white sprinters fades in tailwind conditions, where a flipped effect continues to prevail (on low and high tercile teams). In tough headwind sprints, white sprinters on predominantly white teams show the largest effect yet with a 0.484 s improvement in the post-period, but this point estimate is only statistically significant at the 15 % level after correcting for multiple testing. The 3DiD point estimates for girls in Table 7 are roughly consistent with this pattern, although the headwind effect appears in the low rather than high tercile.

²⁶ Those 3DiD coefficients that are statistically significant by conventional standards become weakly significant after correcting for multiple testing with sharpened q -values of 0.144.

²⁷ As shown in Figure A1 above, the modes of the distribution of Pr(white) are at about 0.06 and 0.93. For the purposes of interpretation of these 3DiD coefficients, however, we use the certainty cases of 0 and 1.

Table 7: 3DiD results for girls 100 m by team average Pr(white) terciles.

	All			Tailwind			Headwind		
	Low	Mid	High	Low	Mid	High	Low	Mid	High
2019	0.040*** (0.013)	0.008 (0.017)	−0.091*** (0.026)	−0.005 (0.027)	0.023 (0.033)	−0.062 (0.049)	0.051 (0.033)	−0.021 (0.041)	−0.071 (0.065)
Post	−0.172*** (0.041)	−0.239*** (0.056)	−0.310*** (0.080)	−0.603*** (0.082)	−0.481*** (0.096)	−0.409*** (0.122)	−0.599*** (0.088)	−0.361*** (0.118)	−0.311* (0.164)
2019 × post	0.032 (0.052)	0.083 (0.075)	0.289*** (0.104)	0.212** (0.103)	0.143 (0.124)	0.459*** (0.164)	0.455*** (0.103)	0.144 (0.159)	0.095 (0.235)
Pr(white)	−0.506*** (0.033)	−0.369*** (0.022)	−0.225*** (0.024)	−0.566*** (0.062)	−0.361*** (0.041)	−0.223*** (0.044)	−0.362*** (0.081)	−0.361*** (0.052)	−0.220*** (0.060)
2019 × Pr(white)	−0.046 (0.047)	0.028 (0.032)	0.083** (0.034)	−0.055 (0.089)	0.078 (0.058)	0.005 (0.065)	−0.155 (0.107)	0.015 (0.073)	−0.012 (0.087)
Post × Pr(white)	−0.275** (0.131)	0.135 (0.096)	0.231** (0.100)	−0.250 (0.231)	0.222 (0.158)	0.140 (0.154)	0.634** (0.261)	0.282 (0.193)	0.407** (0.205)
2019 × post × Pr(white)	0.143 (0.178) [0.591]	−0.115 (0.132) [0.591]	−0.264** (0.132) [0.227]	0.377 (0.317) [0.591]	−0.209 (0.215) [0.591]	−0.241 (0.209) [0.591]	−0.826** (0.344) [0.173]	−0.237 (0.273) [0.591]	−0.131 (0.298) [0.914]
Constant	14.340*** (0.030)	14.210*** (0.030)	14.288*** (0.033)	13.814*** (0.056)	13.462*** (0.053)	13.583*** (0.059)	14.118*** (0.069)	13.835*** (0.068)	13.785*** (0.083)
R-squared	0.47	0.48	0.46	0.49	0.49	0.47	0.48	0.49	0.47
Observations	24,555	24,998	24,059	6359	7402	6831	3934	4493	3868

Data taken from 2018 to 2019 meets in CA. Estimated 3DiD coefficient corresponds to variable 2019 × Post × Pr(white). Observations from 2018 are weighted by entropy balancing to match mean 2019 values of the following covariates for all athletes as well as for those competing in the post-period: average time, standard deviation of time, proportion of races with automatic timing, and the fastest time of the athletes. Included but not reported are grade fixed effects, other control variables from Table 2, and the number of days remaining in the season. Terciles are based on ‘average by team’ predicted Pr(white) (see Figure A1). Standard errors in parentheses. Sharpened q -values for estimated DiD coefficients reported in square brackets. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

6 Discussion

Empirical, often experimental, research provides compelling evidence that beliefs about one’s productivity and likelihood of success can affect effort and realized outcomes. Whether beliefs actually affect performance in practice hinges on the degree to which biophysical constraints bind since slack relative to such constraints is a prerequisite for beliefs to enhance measured productivity. Against this backdrop, we offer a modest contribution using observational data from amateur sports to document the presence of slack in baseline performance and, therefore, the potential for beliefs to affect productivity. We then explore how beliefs shape outcomes using a specific potential self-efficacy shock. Although observational studies like this have the potential virtue of better external validity than artificial settings such as lab experiments, they cannot rule out all confounding factors using exogenous treatment assignment. Our results should be interpreted with this potential virtue and limitation in mind.

The context for this study—the 100 m sprint event in high school track—has a number of useful features

for our analysis. Performance is objectively and precisely measured. Thousands of athletes compete in this event each week, most posting several finishing times in a given track season. Prevailing wind conditions can drive a wedge between physical and psychological determinants of performance, particularly for amateur athletes without years of experience and professional coaching. Outside of a controlled experiment, this setting is likely among the best suited for exploring the role of beliefs on performance.

Our part one results are straightforward and clear: we find evidence of slack in performance relative to the physics that govern drag from wind resistance. This suggests that mindset shapes baseline performance for amateur athletes in distinctive ways relative to elite sprinters. The systematic detrimental effect of headwinds on sprint times beyond increased wind resistance is especially pronounced for girls. This existence of slack in sprinter times is, on average, attributable to mindset effects. This result implies both that self-efficacy beliefs matter in measurable ways in this setting and that a discrete self-efficacy boost could reasonably enhance performance by shrinking the slack in performance.

In part two, we explore these implications using a vicarious injection of self-efficacy that came in the form of Boling's viral 9.98 s 100 m performance in late April 2019. We use a DiD strategy with entropy balancing and multiple testing adjustments to test for a Boling effect, our shorthand term for changes in sprinter times due to athletes' self-efficacy beliefs. The results from this analysis are less clear due to less than perfect causal identification and are more mixed and complex than part one, but nonetheless raise some intriguing potential dimensions of heterogeneity. The conceptual framework that motivates this analysis emphasizes the salience of another person's success as a key factor in the vicarious self-efficacy boost it generates for others. This scope of salience raises a number of facets in our study that likely interact with other forms of mindset heterogeneity, including gender, racial composition of the athlete's own team, and the athlete's own racial background.

Mindset effects in our context are potentially subtle and idiosyncratic, but we can only test for heterogeneous effects using broad categories. In the case of race, which is of particular interest here because of the clear racial framing that often accompanied media coverage of Boling's record sprint, we modify the racial prediction algorithm proposed by Voicu (2018) in order to predict the racial background of athletes in our data. This methodological adaptation may be particularly useful for empirical analyses that use school-level data (e.g., results from high school track and field meets), but these predictions also introduce additional noise and statistical imprecision into our estimates. They also raise ethical issues when the benefit or burden of correct or incorrect racial predictions disproportionately affect specific groups (Lockhart et al. 2023). We consider our use of these predictions to be innocuous since they are not consequential to the individual athletes (e.g., they do not determine eligibility or influence policy or program design). Indeed, we merely use these predictions as an imperfect means of recovering what is often observable by spectators at public track meets, but not recorded in online results repositories. For other applications of this method—adapted for use with school-level data—these potential ethical issues could loom large and must be treated and mitigated with great care.

As a consequence of these heterogeneous results, the empirical evidence we provide suggests that beliefs can indeed shape performance, but these effects do not fit neatly or consistently into a simple model of vicarious self-efficacy. Instead, we find that the Boling effect is a composite of two opposing impacts as enhanced beliefs seem to have improved performance under difficult headwind conditions

for some athletes, but slowed times under favorable tailwind conditions for others. The presence of this flipped effect, which we attribute to amplified overconfidence when a race feels effortless, offers an intriguing and unexpected alternative to the vicarious self-efficacy effect we hypothesized in the conceptual framework. The net effect of these forces determines whether we detect faster or slower times for a specific subset of sprinters. This estimated Boling effect therefore varies according to subtle individual and contextual cues.

Although our empirical setting seems well-suited for addressing our research question, these results offer a complex perspective on the functioning of beliefs and performance. Part of this complexity may emerge from the influence of racial cues, context and identities. The dominant social media narrative about Boling's record-breaking sprint underscored that he is white, but this may have amplified effects on both white and non-white sprinters in different ways—and not just because sprint competitions are zero sum. After all, a 'chip on the shoulder' is a highly effective motivator in athletics and life in general, as few things inspire renewed effort and focus more than feeling slighted or disrespected.

We conclude by considering whether these insights from amateur sprinters may extend beyond the high school track. While the presence of hard biophysical constraints on productivity and performance implies that believing doesn't always make it so, in settings where these constraints are only partially binding—that is, where there is some slack in productivity relative to what is possible given a set of inputs—beliefs can have direct and measurable effects. At least in our context, these effects are complex and reflect subtle contextual and salience cues. Understanding when and how self-efficacy and other beliefs shape outcomes is a worthwhile research goal for economists and social science more broadly. In this sense, this study aims to generate hypotheses that could motivate such follow-on work.

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Appendix: Predicting athlete ethnicity using BIFSG

The method of Voicu (2018) extended the Bayesian Improved Surname Geocoding (BISG) algorithm developed by Elliott et al. (2009) to update prediction probabilities based on first names as well as surnames and location of primary residence (typically, census block-level). The resulting Bayesian Improved Firstname Surname Geocoding (BIFSG) algorithm provided more accurate predictions of an individual's race (Voicu 2018).

We adapt the algorithm used in Voicu (2018) to determine the ethnicity probabilities for each athlete based on their first names, last names, school-level ethnicity probabilities, and school-zip code level ethnicity probability data. For an athlete with surname s who resides in geographic area g , we obtain:

1. the probability of belonging to race or ethnicity r (for each of the six race and ethnicity categories) for a given surname s . $[p(r|s)]^{28}$

²⁸ The probability data is obtained from: <https://www.consumerfinance.gov/data-research/research-reports/using-publicly-available-information-to-proxy-for-unidentified-race-and-ethnicity/>

2. the probability of a person having that first name f , based on his/her race/ethnicity r . $[p(f|r)]^{29}$
3. the proportion of the population of individuals in race/ethnicity r (for each of the six race and ethnicity categories) that lives in geographic area g . $[q(g|r)]$

We then apply Bayes' Theorem to calculate the likelihood that an individual with surname s living in geographic area g belongs to race/ethnicity r using the following equation:

$$\Pr(r|s, f, g) = \frac{p(r|s) * p(f|r) * q(g|r)}{\sum_{r \in R} p(r|s) * p(f|r) * q(g|r)} \quad (4)$$

where R refers to the set of six defined race and ethnicity categories.³⁰ We have to assume that the conditional independence among s , g and f to maintain the statistical validity of the Bayesian updating process i.e., $p(g|r, s) = p(g|r)$ and $p(f|r, s, g) = p(f|r)$.

We apply the school-level racial composition data whenever they are available in step 3 above. Whenever the

²⁹ The probability data is obtained from: Tzioumis, K. Harvard Data-verse <https://doi.org/10.7910/DVN/TYJKEZ> (2018)

³⁰ The six categories defined by the Office of Management and Budget (OMB) for statistical purposes: American Indian or Alaska Native, Asian, Black or African American, Native Hawaiian or Other Pacific Islander, White and Hispanic or Latino.

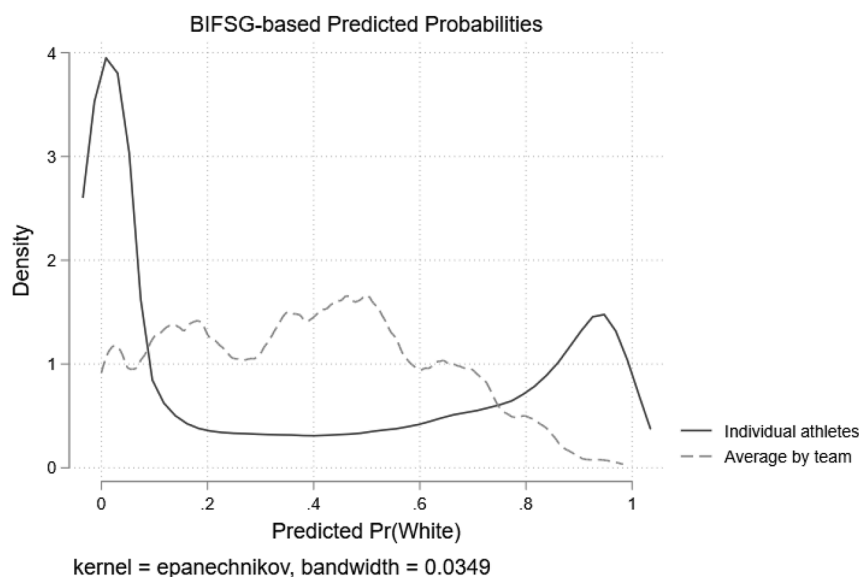


Figure A1: Distribution of BIFSG predicted probabilities that a given boy is white and of team average $\Pr(\text{white})$ for boys in high school 100 m races in CA (Terciles used in subsequent analysis are based on team average $\Pr(\text{white})$).

school-level racial data is not available, we apply the zip-code level ethnicity likelihood obtained from the US Census data. The ethnicity predictions then come in the form of different probabilities for each athlete in the six ethnicity categories.³¹ We take the mean of the probabilities if the athlete's surnames or first names have multiple parts.

Figure A1 shows kernel density distributions of the resulting BIFSG-based predicted probabilities for boys who compete in the 100 m in CA in our sample. The distribution of individual athlete predictions is distinctly bimodal with more boys predicted to be non-white than white (and with fewer boys with $\text{pr}(\text{white})$ in the intermediate range of 0.1–0.8. When we average by team we see that most teams are racially mixed, with average team predictions in the 0–0.7 range. In the analysis, we discretize the average team $\text{Pr}(\text{white})$ distribution into terciles as the basis for team racial composition heterogeneity tests.

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³¹ We calculate the ethnicity likelihood for those athletes' whose surname and first name can be matched to the probability lists used in steps 1 and 2.